Introduction to Economics

Class 10 — Market failures, externalities and public goods

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Plan for today

Warm-up
Since 2005: a permit for every tonne of CO_2
When a deal hits a bystander
Too much glass: the lost triangle, and a tax that fixes it
In pairs: vans in the old town
Too few flu jabs; the noisy terrace
Break
Rules, taxes or permits?
The EU's permits: what happened
In pairs: two power plants
Four kinds of goods; public goods
Fish in a bay; in pairs: lights for a courtyard
In one line; Problem set 2; before Thursday 29 October

Slide 1 Title

Preview version. The full slides and handout are posted after class.

Block 3's third class. Class 9 looked at a market that works as it should and a government that steps in anyway; today the markets themselves get it wrong. First, deals that hurt or help people who were not part of them, and the tools that can put that right: a tax, a bargain, a rule, permits. After the break, goods nobody can be kept from using, which a market provides too little of or uses up too fast. The thread is a real price on CO_2: since 2005 every large plant in the EU must hand in a permit for each tonne, and the question is what that did.

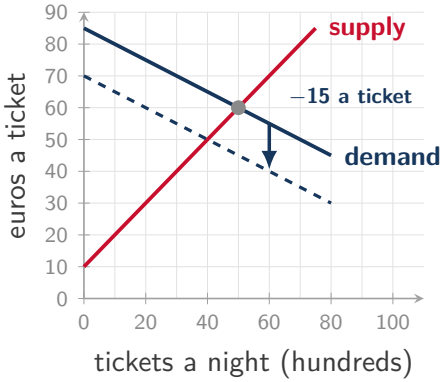
Slide 2 Plan for today

The class at a glance. Up to the break the market makes too much or too little of something because a bystander is left out of the price: a glassworks' smoke, delivery vans, flu jabs, a noisy terrace. After the break, what a government can do when the people involved cannot settle it among themselves, a rule, a tax or tradable permits, and then the goods that no market handles well at all, from Christmas lights to fish in a bay.

Two halves of fifty minutes with a ten-minute break; Problem set 2 is handed out at the end.

Warm-up: a tax paid by the buyers — who pays what?

Flamenco shows: a tax of 15 a ticket, paid by the buyers



1. Tickets now, and what venues keep?
2. What does a buyer pay in all?
3. Who bears more of the 15, and why?

In pairs, two minutes; then hands per item.

Slide 3 Warm-up: a tax paid by the buyers — who pays what?

A warm-up on Class 8's tax collected from the buyers, read on a figure. Before class, find where supply meets the dashed line and ask what that price is: what the venues keep, or what a buyer pays in all? Then compare both prices with the price before the tax, and think about what decides which side bears more.

Revealed in class.

Warm-up: consumer surplus, buyer by buyer

Four friends, one tandem paragliding flight each at most

The most each would pay: 150, 125, 100, 70 euros. A flight costs 90.

1. Who buys a flight?
2. Their consumer surplus?
3. Does the friend who would pay 70 count?

In pairs, one minute; then hands per item.

Slide 4 Warm-up: consumer surplus, buyer by buyer

The second warm-up is Class 8's consumer surplus, counted buyer by buyer. Before class, decide who buys at the price of a flight, work out what each buyer gains, and ask whether everyone in the group counts.

Revealed in class.

Since 2005: one permit for every tonne of CO_2

a power station or a steelworks in the EU → **hands in one permit for every tonne of CO_2 it emits**

The EU limits the number of permits; firms trade them.
In 2024 one cost about 65 euros.

Did emissions fall — and by how much?

EU Emissions Trading System (ETS).
European Commission, Carbon market report COM(2025) 735, 3 December 2025.

Slide 5 Since 2005: one permit for every tonne of CO_2

The hook, from the European Commission's own report. Since 2005 every large power station and factory in the EU, since 2012 also flights within Europe and since 2024 shipping, together about two fifths of the EU's emissions (the Commission's figure for 2024), must hand in one permit for each tonne of CO_2 it emits in a year; the EU fixes how many permits exist and issues fewer each year, and firms that need more buy them from firms that need fewer. The slide asks what twenty years of that did to emissions. Keep the question open: after the break the deck shows what a cap does, and then the data. The idea behind the whole class is already here: one plant's CO_2 lands on everyone.

An externality: a deal between two that hits a third

buyer and seller: a deal both want → **a bystander: a cost or a benefit, unpaid**

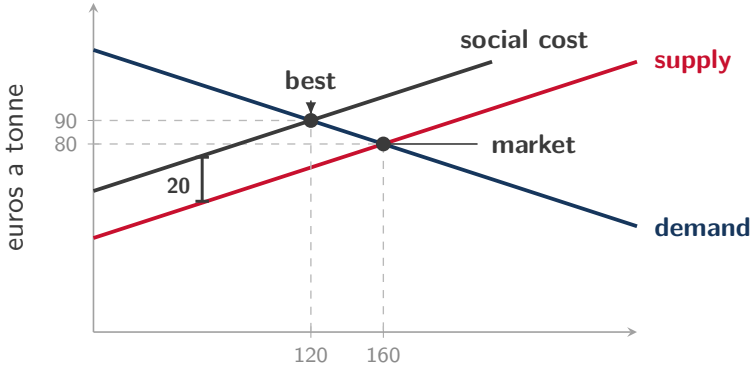
Negative: a bar's terrace keeps the street awake
Positive: your flu jab protects your flatmates

The price counts the deal, not the bystander.

Slide 6 An externality: a deal between two that hits a third

The definition in words: an externality is the effect of a decision on a bystander who is neither paid for it nor pays for it. A harm is a negative externality, a benefit a positive one. The market price weighs only the two sides of the deal, the buyer's value and the seller's cost, so it can push the quantity too high when bystanders are hurt and too low when they gain; the next slides put numbers on both. Not every effect on others counts: a price that rises because more people want the good is felt by other buyers, but it is paid for through the market, so it is no externality.

Glass: the market makes 160; counting the smoke, 120 is best

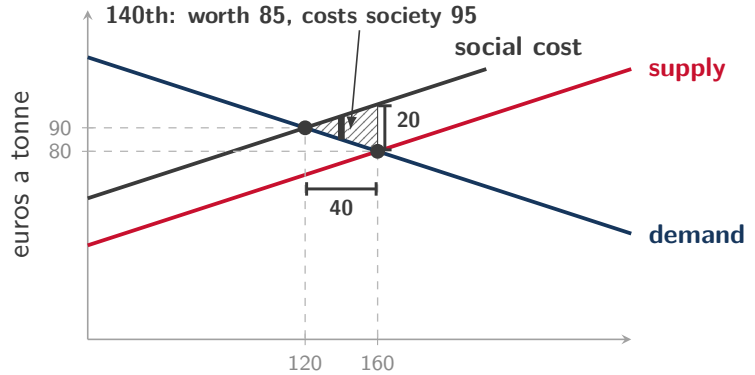


Social cost = the works' own cost + 20 euros of smoke a tonne.

Slide 7 Glass: the market makes 160; counting the smoke, 120 is best

The core picture, built in three steps. Demand is what builders and bottlers would pay for each tonne of glass; supply is the works' own cost of making it, and the market settles where the two meet. The smoke adds a cost nobody in the deal pays, so the cost to society of a tonne is the works' cost plus the harm, drawn as a line the same distance above supply everywhere. The best quantity is where the value of the last tonne just covers its full cost, and it is smaller than the market's but not zero: glass is worth having, so the best amount still makes some smoke. The social cost is not the harm alone, which is a common slip.

Tonnes 120 to 160 cost society more than they are worth: 400 thousand euros a year

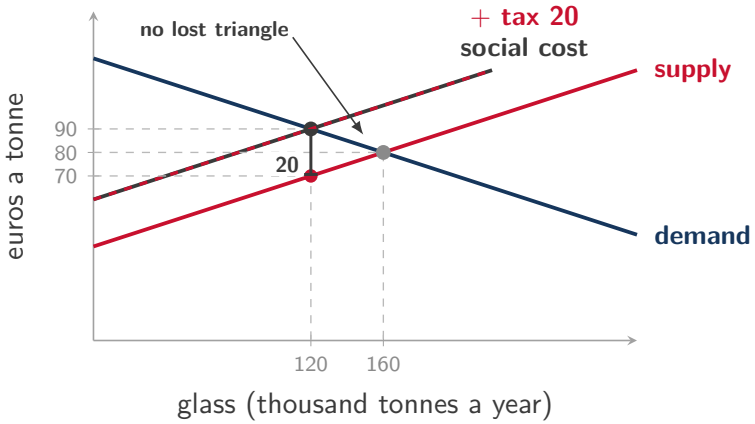


Lost: ½ × 20 × 40 = 400 thousand euros a year — Class 9's triangle, now from too much.

Slide 8 Tonnes 120 to 160 cost society more than they are worth: 400 thousand euros a year

Why the extra tonnes are a loss and not just a nuisance. Each tonne between 120 and 160 is worth less to its buyer than it costs the works and the town together, yet the market makes it, because the two in the deal only see their own gain. The losses start at nothing at 120 and grow to the full harm per tonne at 160, which is why they form a triangle, and half of base times height is the same formula as Class 9's. The difference is the direction: there a tax stopped deals worth making, here the market makes deals not worth making. The loss is not the harm on all 160 tonnes; the first 120 are worth more than their full cost.

A tax of 20 a tonne: the market stops at 120, and the lost triangle is gone



Pigovian tax: a tax that corrects a negative externality; set at the external cost (20).

Slide 9 A tax of 20 a tonne: the market stops at 120, and the lost triangle is gone

The fix, and the reason it is exactly 20. With the tax the works pay their own cost plus the harm, so their supply curve moves up by 20 and lands on the social cost line: the deal now counts the bystander, which the reading calls internalising the externality. The market then finds 120 on its own, buyers pay 90 and the works keep 70, and the town collects the tax on every tonne still made. This is a Pigovian, or corrective, tax: a tax that corrects a negative externality, here set at the external cost per unit. In Class 9 a tax on coffee created a lost triangle because that market had nothing wrong with it; here the market was wrong to begin with, and the tax removes the loss.

In pairs, four minutes: vans in the old town

Deliveries in the old town

deliveries a day	last one is worth	last one costs the firm
160	22	14
200	20	16
240	18	18

Each delivery's fumes and noise cost residents 4 euros.

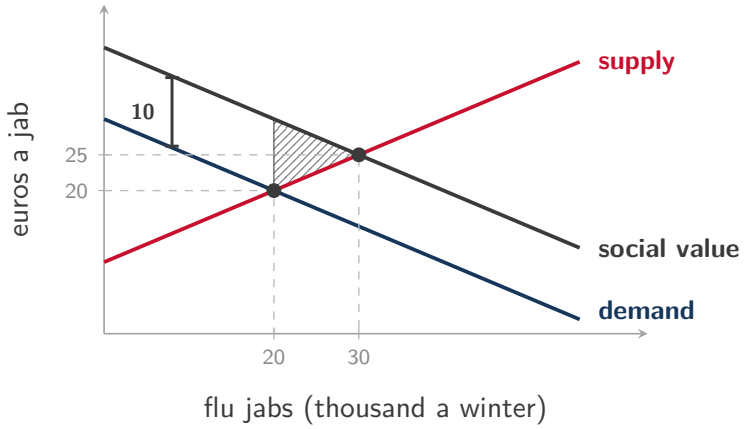
1. How many deliveries does the market make?
2. How many are best, counting the residents?
3. Which tax gets the market there?
4. The lost surplus from making 240, not 200?

Slide 10 In pairs, four minutes: vans in the old town

Exercise 1 is slides 7 to 9 on a table instead of a graph. Before class, find the row where the shops' worth equals the firm's own cost, then the row where it equals that cost plus the residents' 4, and ask which tax would make the firm stop there by itself. For the lost surplus, decide which deliveries form the base of the triangle and what its height is.

Revealed in class.

Flu jabs protect others too: the market gives 20 thousand, society wants 30



Too few: a subsidy of 10 a jab gets the market to 30.

Slide 11 Flu jabs protect others too: the market gives 20 thousand, society wants 30

The mirror image of the glass. Your demand counts the protection you get; the jab also protects the people around you, so society's value curve sits above demand by the benefit to others, and the best quantity is where that curve meets the cost of a jab. The market stops short of it: the jabs between the two quantities are worth more to everyone than they cost and are never given, and their lost triangle is slide 8's formula again, half of 10 times the 10 thousand jabs not given. The fix mirrors the tax: a subsidy equal to the benefit to others, which is why health systems give flu jabs free to older people and risk groups. In Class 8 a subsidy created a lost triangle; here it removes one.

The noisy terrace: does it matter who holds the right?

A bar's terrace, open late

The bar gains 200 a night · the noise costs the 30 neighbours 300 a night

1. The bar has the right to stay open. What happens?
2. The neighbours have the right to quiet. Now?
3. The bar gains 400 a night instead: open or closed?

In pairs, two minutes; then hands.

Slide 12 The noisy terrace: does it matter who holds the right?

The terrace promised in Class 9, and a way out that needs no state. Before class, compare what the bar gains with what the neighbours lose, then ask what each side would be willing to pay the other in each case and whether the terrace ends up open or closed; then ask what changes when the bar's gain is larger.

Revealed in class.

A tax on plastic cups: which curve moves? (optional)

A tax of 0.10 on every plastic cup, paid by the makers

1. Which curve shifts, which way?
2. Cups get dearer: does demand shift too?
3. Reusable cups get cheaper. Now?

Slide 13 A tax on plastic cups: which curve moves? (optional)

An optional exercise on Class 5's shift question, tied to slide 9's tax. Before class, ask for each event whether it changes what makers would supply at every price, what buyers would want at every price, or only the price itself; then decide which curve, if any, moves and which way.

Revealed in class.

Break

Ten minutes

Slide 14 Break

Ten minutes. After the break: why thirty neighbours rarely strike the terrace bargain, and what a council can do instead.

Thirty neighbours rarely bargain: so the council sets a rule or a price

- Thirty neighbours: who organises them, who pays the lawyer?
- Each hopes the others will pay
- Each side holds out for a better deal
- So the council steps in: a closing hour, or a charge

Slide 15 Thirty neighbours rarely bargain: so the council sets a rule or a price

Why slide 12's bargain stays on paper when the bystanders are many. Getting thirty people into one room, agreeing who pays what and paying a lawyer are transaction costs that can exceed what the deal is worth; each neighbour's quiet also benefits all the others, so each waits for someone else to pay, a problem that returns under its own name with the Christmas lights; and once both sides hold out for a better split, the deal can fall apart even though one exists. When private bargaining fails, the state can act for the many, either with a rule, such as a closing hour or an emissions limit, or with a price, such as a tax on each unit of harm. The next slides compare the two.

Cutting 4 thousand tonnes costs 100 under a rule, 40 under a tax

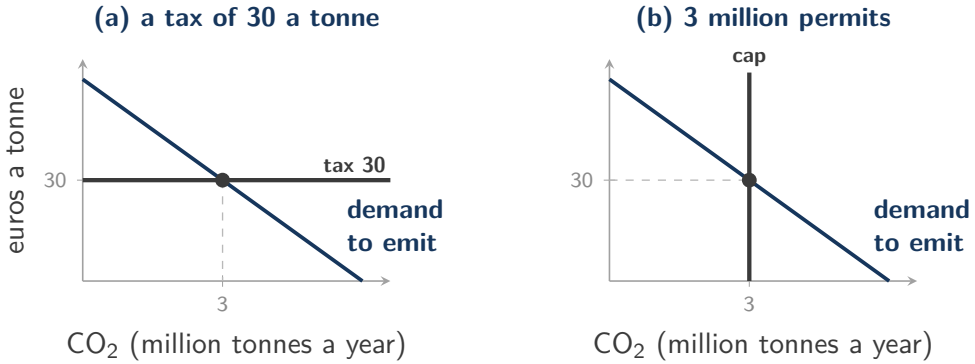
	Works A	**Works B**	**cost of cutting** (thousand euros)
CO_2 now (thousand tonnes a year)	4	4	
cutting a tonne costs (euros)	10	40	
Rule: each cuts 2	2×10	2×40	**100**
Tax 25 a tonne emitted	cuts 4	cuts none	$\mathbf{4 \times 10 = 40}$

The cheap cutter does the cutting.

Slide 16 Cutting 4 thousand tonnes costs 100 under a rule, 40 under a tax

Why a price beats a rule for the same cut. The rule makes each works cut the same amount regardless of what cutting costs it, so the works for which cutting is dearest does as much of it as the cheap one. Under the tax each works compares cutting with paying: A cuts because cutting is cheaper than the tax, B pays because paying is cheaper than cutting, and the whole cut lands on the cheap cutter. B's tax payment is not part of the cost of the cut; it goes to the town, which can lower other taxes with it. The town did not need to know which works cuts cheaply, since the tax sorts that out, though it did need a tax somewhere between the two cutting costs, and because the tax keeps charging for every tonne still emitted it also rewards a cleaner technology, where a rule stops at the limit; whether equal cuts are fairer is a different question from whether they are cheaper.

A tax fixes the price of emitting; a permit cap fixes the quantity



Same point: a tax picks the price, a cap picks the quantity.

Slide 17 A tax fixes the price of emitting; a permit cap fixes the quantity

The two prices on the same picture. The line is every firm's tonnes lined up from the dearest to cut to the cheapest, so its height is what a firm would pay to emit one more tonne, which is what cutting that tonne would cost it. With a tax, each firm emits the tonnes that cost more than the tax to cut and cuts the rest, so the quantity follows from the price; with permits the state fixes the quantity and trading sets the price, since firms with dear cuts buy from firms with cheap ones, and a permit ends up worth exactly what the last cut costs. Set right, the two land on the same point; they differ in what is certain, the quantity or the cost to firms. Even a firm given its permits for free pays to pollute, because it could have sold them: Class 2's opportunity cost.

The EU's permits: what happened

Revealed in class

Slide 17: a cap fixes the quantity. Guess by how much twenty years of a shrinking cap cut emissions.

Slide 18 The EU's permits: what happened

Revealed in class.

Beforehand: take slide 5's question to slide 17. A cap fixes the number of tonnes and lets the price do the sorting; ask what twenty years of a cap that shrinks every year should do to the emissions of the plants under it, and what else, besides the permit price, might have moved them over the same years.

In pairs, four minutes: two power plants, a rule, a tax, permits

	Plant N	Plant S
CO_2 now (thousand tonnes a year)	6	6
cutting a tonne costs (euros)	20	50

Goal: 6 thousand tonnes less in all.
Costs in thousand euros.

1. Rule: each cuts 3. Cost?
2. Tax 30 a tonne emitted: who cuts?
3. The cost of the cut with the tax?
4. Tradable permits to emit 3 each: who sells?

Slide 19 In pairs, four minutes: two power plants, a rule, a tax, permits

Exercise 2 is slide 16 with a third policy, tradable permits. Before class, cost the rule row by row, then ask for each plant whether cutting a tonne or paying the tax on it is cheaper, and add up only the cutting. For the permits, ask what each plant would pay or accept for one, and which way they would move.

Revealed in class.

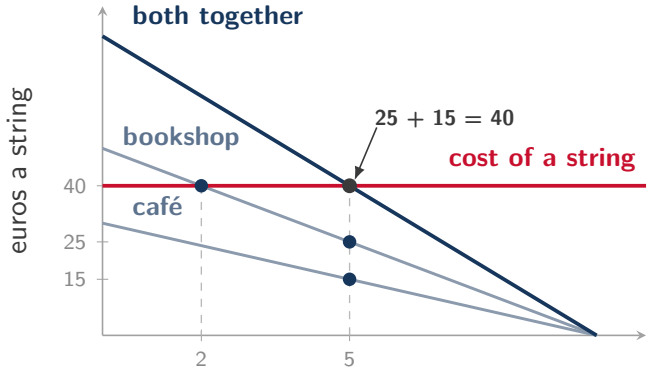
Two questions sort goods into four kinds

	excludable (non-payers kept out)	**not excludable** (anyone can use it)
rival (my use leaves less for you)	**private good** a cup of coffee	**common resource** fish in a bay
not rival (it does not)	**club good** a toll motorway, no jams	**public good** a street's Christmas lights

Slide 20 Two questions sort goods into four kinds

The other side of market failure starts with a sorting rule. Excludable means those who do not pay can be kept out; rival means my use leaves less for you. Markets handle goods that are both, because the seller can charge and each unit goes to one buyer; each of the other three boxes fails in its own way, and the next slides take the two that are open to everyone. The same good can change box: a free road is a public good at four in the morning and a common resource at rush hour. A public good is not simply anything the state provides; it is defined by the two questions, whoever supplies it.

A public good: add what each would pay for each string — the street wants 5



Both get the same lights: add what each would pay.

Slide 21 A public good: add what each would pay for each string — the street wants 5

Why the market lights too little of the street. Each shop alone hangs strings while a string is worth the cost to it, and the café never starts. But every string lights the whole street, so a string's value to the street is what the bookshop would pay plus what the café would pay for that same string, and that sum covers the cost of a string further out than either shop alone. The gap is the free rider: the café enjoys the bookshop's strings without paying and the bookshop ignores the café's value, so strings worth hanging are not hung. The fix is to make everyone pay a share, by a club or by taxes, as a council does with fireworks.

A free bay fills with boats: each new boat takes fish from the others

Boats in a bay

boats	catch in all	per boat
	(thousand euros of fish a week)	
1	12	12
2	22	11
3	30	10
4	36	9
5	40	8
6	42	7

A boat's week costs 7.

4th boat: catches 9, but the bay gains only 6

Open bay: six boats, each just covers its 7

Best: three boats; $30 - 21 = 9$ a week left

Rival, open to all: used too much.

Slide 22 A free bay fills with boats: each new boat takes fish from the others

The other open-access failure, rival this time. A skipper sees his own catch, but the bay's total catch rises by less than that, because part of what the new boat lands is fish the boats already there would have caught: that shortfall is the externality inside a common resource. Since nobody can be kept out, boats keep coming while a boat's own catch covers its cost, until the whole catch is eaten up by costs and nothing is left over. For the bay, a boat is worth adding only while it adds at least its cost to the total catch, which stops much earlier. This is the tragedy of the commons; the overuse comes from open access and rivalry, not from greed, and the fixes all restore one or the other: licences or quotas, a charge per boat, or an owner for the bay.

In pairs, three minutes: lights for a shared courtyard

Two flats share a courtyard; a string costs 25

	1st	2nd	3rd	4th string
Flat A would pay	30	20	10	0
Flat B would pay	20	15	10	5

1. Flat A alone: how many strings?
2. Flat B alone?
3. Both together?
4. Why might they end up with fewer?

Slide 23 In pairs, three minutes: lights for a shared courtyard

Exercise 3 is slide 21 on a table. Before class, decide for each flat alone which strings are worth their cost to it, then ask what the two flats together would pay for the same string and compare that with the cost, string by string. Then think about why they might still end up with fewer than that.

Revealed in class.

What does a morning's beach clean-up cost you? (optional)

Saturday: 4 hours cleaning the beach

You could earn 11 an hour instead · the bus to the beach costs 3
Your monthly gym fee is already paid

1. Your opportunity cost of the morning?
2. Is the gym fee part of it?
3. Why do too few people volunteer?

Slide 24 What does a morning's beach clean-up cost you? (optional)

An optional exercise on Class 2's opportunity cost, tied to slide 21. Before class, list what the morning on the beach actually makes you give up, and ask of each item on the slide whether it changes if you stay at home; then ask who gets the clean beach and who paid for it.

Revealed in class.

A tax on petrol: positive or normative? (optional)

A tax on petrol: three statements

1. "A tax of 0.20 a litre cuts driving more than one of 0.10."
2. "Cleaner air is worth dearer driving."
3. Both want cleaner air. B: "Car-free zones instead: drivers barely react to prices." B's reason: values or a prediction?

Slide 25 A tax on petrol: positive or normative? (optional)

An optional exercise on Class 4's distinction, on the taxes of slides 9, 16 and 17. Before class, ask of each statement whether evidence could in principle show it to be wrong; for the third, ask whether B disagrees with the goal or with a forecast.

Revealed in class.

A cost on a bystander: the market makes too much; a tax equal to that cost fixes it.

Open to all: a public good gets too little, a common resource is used too much.

A price lets the cheapest cutter do the cutting.
Thursday: what it costs a firm to make things.

Slide 26 In one line

The class in four lines, to read slowly: the first is slides 6 to 10, the second slides 20 to 23, the third slides 16 to 19, and the fourth names Thursday.

Before Thursday 29 October

- Thursday: production and costs
- Read Mankiw & Taylor: Ch. 5, pp. 105–122
- Problem set 2 (Block 3): handed out today; due Monday 2 November, on paper, in class
- Practise in Sextant: the units the course page lists



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Slide 27 Before Thursday 29 October

Thursday 29 October: what it costs a firm to make things, in the short run and the long run: production and costs. Reading: Mankiw & Taylor chapter 5, pages 105 to 122. Problem set 2, on Block 3, is handed out in class and is due on Monday 2 November, on paper, in class. Practise in Sextant with the units the course page lists.

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